

Meiru Zhang

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Research areas: Large Language Model, Retrieval-Augmented Generation, Agent, Personalization, Event Extraction and Forecasting, Neurosymbolic AI, Event-centric World Model

EDUCATION

- University of Cambridge**, PhD in Language Technology Lab Oct 2021 – Aug 2026
• *Supervisor:* Prof. Nigel Collier | **Gates Cambridge Scholarship** | *Research:* Event Extraction/Forecasting, RAG, LLM Agents.
- Imperial College London**, MSc in Computing (Machine Learning) Sept 2018 – Sept 2019
• **Distinction** | *Research:* Personalization, Active Learning, Reinforcement Learning | *Thesis:* **Multimodal Active Learning from Human Data** (learn-to-learn; data-scarce engagement prediction in autism sessions).
- University College London**, BEng in Electronic and Electrical Engineering Sept 2015 – Jun 2018
• **GPA:** 81 (Top 5%) | **Dean’s List** | Goldsmith Sessional Prize (2016, 2017)
• *Project:* **Big Data Applied on Registered Patents** (TF-IDF + ML to forecast fuel-cell patent growth).

PUBLICATIONS

- **Failure Modes in Multi-Hop QA: The Weakest Link Law and the Recognition Bottleneck** Under review. Meiru Zhang, Zaiqiao Meng, Nigel Collier.
- **DRIFT: Decompose, Retrieve, Illustrate, then Formalize Theorems** Accepted to ICLR 2026. Meiru Zhang, Philipp Borchert, Milan Gritta, and Gerasimos Lampouras.
- **The Power of Simplicity in LLM-Based Event Forecasting** In Proceedings of the First Workshop for Research on Agent Language Models (REALM) at ACL 2025. Meiru Zhang, Auss Abbood, Zaiqiao Meng, and Nigel Collier.
- **Can We Instruct LLMs to Compensate for Position Bias?** In Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing. Meiru Zhang, Zaiqiao Meng, and Nigel Collier.
- **COFFEE: A Contrastive Oracle-Free Framework for Event Extraction.** In Proceedings of the First Workshop on Matching From Unstructured and Structured Data (MATCHING 2023, ACL). Meiru Zhang, Yixuan Su, Zaiqiao Meng, Zihao Fu, and Nigel Collier.
- **Multi-modal Active Learning From Human Data: A Deep Reinforcement Learning Approach.** In 2019 International Conference on Multimodal Interaction (ICMI '19). Ognjen Rudovic, Meiru Zhang, Bjorn Schuller, and Rosalind Picard. 2019.

RESEARCH EXPERIENCE

Failure Modes in Multi-Hop QA: The Weakest Link Law and the Recognition Bottleneck Aug 2025 – Jan 2026

- **Problem:** Multi-hop QA fails in long contexts due to position bias.
- **Action:** Proposed *Multi-Focus Attention Instruction* (MFAI); evaluated 5 LLMs on MuSiQue and NeoQA.
- **Result:** Identified recognition failure as the main bottleneck; proposed the “Weakest Link Law”; matched MFAI improves accuracy by up to 11.5%; System-2 models match gold-only baselines in noisy long-context inputs.

The Power of Simplicity in LLM-Based Event Forecasting July 2024 – Mar 2025

- **Problem:** LLM-based event forecasting is often expensive due to iterative agentic reasoning (e.g., ReAct) and noisy, unstructured context.
- **Action:** Replaced multi-step agents with a single-step RAG pipeline; designed compact, structured retrieval formats (statistical distributions, event code lists) and ran controlled evaluations.
- **Result:** Showed structured evidence beats raw headlines at ~10% of ReAct inference cost.

Can We Instruct LLMs to Compensate for Position Bias?

Dec 2023 – May 2024

- **Problem:** In-context learning is unstable under position bias.
- **Action:** Proposed **Attention Instruction** to steer attention; tested multiple designs.
- **Result:** Improved controllability by reweighting attention to target prompt regions via context intervention.

Contrastive Oracle-free Framework for Event Extraction (COFFEE)

May 2022 – Jan 2023

- **Problem:** Event extraction pipelines often rely on oracle trigger annotations and brittle assumptions that do not generalize.
- **Action:** Introduced the Oracle-Free Event Extraction (OFEE) setting and developed the COFFEE framework; led method design and benchmarking against prior approaches.
- **Result:** Enabled event extraction without oracle triggers and provided a stronger, more realistic evaluation protocol for downstream systems.

Multi-modal Active Learning from Human Data

Mar 2019 – Aug 2019

- **Problem:** Labeling multimodal human interaction data is costly; models need to learn efficiently from limited annotations.
- **Action:** Built multimodal user-state models and implemented a DQN-based active learning agent to select the most informative samples for labeling.
- **Result:** Reduced annotation waste by actively prioritizing high-information training examples for multi-class classification.

WORK EXPERIENCE

Huawei Noah's Ark Lab, London, UK, Full-time Research Intern in NLP

Feb 2025 – Sept 2025

- **Situation/Task:** Improve retrieval-augmented theorem autoformalization with LLMs, addressing under-specified retrieval queries and weak guidance on how to use retrieved premises.
- **Action:** Built a retrieval-augmented autoformalization pipeline (DRIFT): decompose queries, retrieve premises and illustrative examples with a finetuned BGE-M3 retriever, and then add to context to assist the autoformalization
- **Result:** DRIFT achieves an absolute improvement of 13.34, 2.08, and 7.74 points over the baseline F1, enabling downstream autoformalization on OOD ConNF benchmark to improve by 42.25% and 37.14% absolute points on BEq+@10; **paper accepted to ICLR 2026**.

Antobot, Computer Vision Engineer

Oct 2018 – Sept 2021

- **Situation/Task:** Deploy robust fruit detection, tracking, and yield estimation on edge devices for field robotics.
- **Action:** Delivered an end-to-end perception stack by building an instance-segmentation dataset (video curation, guidelines, label QA), training detection/tracking models (MobileNet, YOLOv5), and developing regression modules for yield and fruit size estimation.
- **Result:** Delivered an on-device perception pipeline by integrating models with TensorRT and ONNX for production edge deployment.

Westlake University, Research Assistant

Sept 2020 – Jan 2021

- **Situation/Task:** Improve fast adaptation in non-stationary, low-data settings in the Machine Intelligence Lab (MiLab).
- **Action:** Investigated meta-learning and “learning to active learn” strategies; designed experiments to evaluate rapid adaptation under distribution shift.
- **Result:** Produced actionable insights on combining meta-learning with active data acquisition for more sample-efficient adaptation.

LANGUAGE & SKILLS

Languages: Chinese (Native), English (Fluent), German (A2) | **Programming Language:** Python, Java, C

Frameworks/Tools: PyTorch, TensorFlow, Transformers, OpenCV, vLLM, Weights and Biases.